# Requirements

**Team:** CS.054 FracFeedExtractor
**Members:** Bradley Rule, Raymond Cen, Sean Clayton, Zahra Zahir Ahmed Alsulaimawi
**Date:** Oct 26th, 2025

## Problem Statement & Scope

**Problem Statement:**
Understanding predator-prey interactions is essential for the biology department's research, but the key metric of the fraction of feeding individuals is often underused because it is buried in unstructured scientific papers. The FracFeedExtractor system will automate the extraction and identification of predator diet surveys using LLMs. It will classify documents for relevance and extract unstructured data, such as the count of empty vs. non-empty stomachs, species, location, and the year of the study to expand the department's predator-prey database.
**Primary Users:**
- Ecological researchers at OSU validating the fraction of feeding predators.
- Data scientists maintaining or extending the FracFeed database.

**Secondary Users:**
- Developers expanding the system and adapting to new data formats or models.
- Biology students at OSU interested in predator-prey relationships.

**Context of Use:**
The system will operate as a Python command-line pipeline that is capable of processing individual PDFs or folders of PDFs. It will run in research computing environments such as local workstations or university servers. PDFs are passed to the system, text is preprocessed and extracted, and passed to an LLM to extract key metrics.
**System Boundaries:**

**In Scope:**
- PDF parsing and text extraction.
- Document classification and key data extraction.
- Trained model specifically for predator-prey publications.
- Structured export with provenance and confidence value.
- Reproducible training and evaluation pipelines.

**Out of Scope (Non-Goal):**
- Performing ecological modeling or analysis beyond data extraction.

**Measurable Success Criteria:**
1. **Extraction Accuracy:** ≥90% precision and ≥85% recall on labeled test PDFs.
2. **Processing Efficiency:** Typical 20-page PDF processed in under 10 seconds.
3. **Usability:** ≥80% of users can run the tool successfully without assistance.
4. **Classification Confidence Reporting:** The system shall output a confidence score (0-1) for each document classification with an accuracy of ±0.05 compared to validation data.

## **Consolidated Requirements**

| ID | Requirement | Acceptance Criteria | Priority |
|---|---|---|---|
| REQ-001 | The system should classify PDF as useful or not useful for predator diet data | Given a PDF, when classification modules runs then the output shall include a header whether the PDF is useful or not | Must |
| REQ-002 | The system shall extract key predator data( ex. Species, count of empty vs non empty stomach, location and year) | Given useful PDF when the extraction all required fields shall present the output of the data ( ex. Species, count of empty vs non empty stomach, location and year) | Must |
| REQ-003 | The systems shall correctly interpret tables and figures, ensuring the LLM does not hallucinate or takes incorrect values. | Given PDF containing tables or figures, when the table detection process runs, then all values shall match and be flagged if uncertain. | Must |
| REQ-004 | The system shall automatically apply OCR when a PDF page has no readable text layer. | Given an image based or scanned PDF when the system detects no text then OCR shall automatically extract readable text for the further processing. | Must |
| REQ-005 | The system shall track and display accuracy and page numbers for the extracted. | Given an extraction run when the results are outputted then each extracted item shall include page | Must |

| | | | |
|---|---|---|---|
| | | number to make sure of accuracy. | |
| REQ-006 | The model used for classification and data extraction should be fine tuned for predator-diet surveys. | Given a labeled training dataset when model fine-tuning is completed then the systems classification accuracy shall reach at least 95% and data extraction should reach accuracy 95% when testing valid PDF. | Should |
| REQ-007 | The system shall analyze full PDF inputs to ensure no relevant data sections (tables, captions, appendices, etc.) are missed. | Given a PDF containing multiple sections when extraction module runs then all relevant tables,captions and appendices are scanned and included in the output without anything missed. | Should |
| REQ-008 | The system shall log all processed pages, errors, and any summary statistics. | Given a PDF when the pipeline completes then the system creates a log file that lists total pages processed, any failed pages, errors and success. | Must |
| REQ-009 | The system shall support both single PDF uploads and folders of PDFs. | Given either one PDF or a folder of PDFs when the upload command is executed then the system processes all valid files without user reconfiguration. | Should |
| REQ-010 | The system should be able to convert any | Given any valid PDF file, when the conversion is | Must |

| | | | |
|---|---|---|---|
| | PDF to a raw text format | triggered then a clean readable text version of the document is generated for downstream processing. | |

## Prioritization Method & Results

Our team used the MoSCoW method to prioritize requirements, classifying each as Must Have, Should Have, and Could Have. The requirements were classified by what we felt were important to our project partner, technical feasibility, and project timeline.

- Must Have requirements represent core functionality that are essential for meeting the project goals. In this case, accurate classification, extraction of key metrics, and using OCR as a fallback for image based PDFs are all Must Have requirements.
- Should Have requirements add significant value, but could possibly be avoided if needed due to time or budget constraints.
- Could Have items are lower priority enhancements that add minimal value, but would be beneficial if we have time for them.

**Tie Breaking Criteria:**
When two items received similar priority ratings, we felt that requirements that bring more value to the project partner, or overall project goal, as well as requirements that mitigate risk (such as technical or data risk) should be prioritized. We felt this was the best way to break tiebreakers as at the end of the day, the project is for the project partner so their requirements should be our first priority.

**Key Dependencies:**

- REQ-001 (classification) depends on REQ-010 (PDF to text conversion) and REQ-004 (OCR fallback) as accurate text extraction is required before any classification can occur.
- REQ-002 (data extraction) depends on REQ-003 (table detection) and REQ-007 (full PDF analysis) to ensure complete and accurate extraction of relevant predator diet data.
- REQ-005 (accuracy) depends on both REQ-001 and REQ-002 as accuracy can only be calculated after classification and extraction steps are functional.
- REQ-008 (logging and statistics) depend on all core processing requirements, like REQ-001 through REQ-007, as logs and summaries are generated based on outputs from each stage.
- REQ-009 (batch processing) depends on the stability of single PDF processing (REQ-003, REQ-004, and REQ-007) to ensure consistent behavior when processing multiple files.

**Ethics, Risks, and Constraints Traceability**

| Related Requirements | Concerns and Risks | Mitigation/Acceptance Notes |
|---|---|---|
| REQ-001, REQ-002, REQ-006 | Concern for intellectual property and the legal use of certain papers or materials | Only process legally obtained papers/PDFs through institutions or open access sources. Include disclaimers and check for licensing before processing. |
| REQ-003, REQ-005 | LLM can infer inaccurate values or data and mislead researchers with corrupted data. | Implement a confidence score and flag low scores for manual review. |
| REQ-002, REQ-006 | Inaccurate or non reproducible extraction of data and values can weaken the trust of the research on the LLM | Maintain a reproducible training and evaluation pipeline. Version datasets, models and configuration. |
| REQ-004, REQ-008 | Inaccessible tools or not enough documentation can exclude less technical researchers and prevent researchers from wanting to use the tool | Provide clear README, documentation, and logs. Test with intended researchers in mind. |
| REQ-004 | Inefficient OCR can inhibit scalability and waste computing resources | Track performance metrics for every PDF. Optimize OCR calls and encourage efficient use of research computing resources |
| REQ-006 | If the training data is skewed then the extracted results can misrepresent global predator and prey patters | Use diverse training sets. Retrain and validate models against underrepresented data categories. Document any known limitations |

**Merge Methodology & Change References**

When merging we first looked at which items overlapped each other and which items were similar in intent. Then we merged all of those into a single requirement and assigned a unique ID to them. Conflicts or priorities were resolved by consensus with the project partner's values and technical implementation/output in mind. Each requirement is mapped to at least one team member's idea. This ensures every team member's input was considered and represented in a consolidated set.

| **Team REQ ID** | **Personal** | **References** |
|---|---|---|
| REQ-001 | Sean, Raymond, Bradley, Zahra | Sean: classify publications as useful/not useful; Raymond: classify papers as useful/not useful; Bradley: classify PDFs with predator data; Zahra: indicate whether a PDF is useful or not. |
| REQ-002 | Sean, Raymond, Bradley, Zahra | Sean: extract predator diet data; Raymond: extract specific data from useful papers; Bradley: extract proportion of feeding predators; Zahra: extract data tables from PDFs. |
| REQ-003 | Bradley, Zahra | Bradley: process figures/models to extract data; Zahra: track accuracy, prevent hallucination errors, ensure correctness in tables. |
| REQ-004 | Sean, Bradley | Sean: apply OCR to scanned PDFs; Bradley: convert PDF to raw text for analysis. |
| REQ-005 | Raymond, Zahra, Sean | Raymond: confidence scores and accuracy tracking; Zahra: track accuracy and show page numbers; Sean: measurable accuracy target (≥90%). |

| | | |
|---|---|---|
| REQ-006 | Sean, Raymond, Bradley, Zahra | Sean: academic data licensing compliance; Raymond: redact personal info for privacy; Bradley: adhere to research data regulations and use ML/LLMs; Zahra: follow guidelines and maintain consistency. |
| REQ-007 | Bradley, Zahra, Raymond | Bradley: process large datasets and run unsupervised; Zahra: extract from multiple PDF sections; Raymond: measure runtime and performance efficiency. |
| REQ-008 | Sean, Raymond, Zahra, Bradley | Raymond: logging of processed PDFs and results; Sean: include testing and documentation; Zahra: display error logs and remove duplicates; Bradley: structured spreadsheet output. |
| REQ-009 | Sean, Raymond, Zahra, Bradley | Raymond: upload single/multiple PDFs and summarize results; Sean: batch processing for scalability; Bradley: process large sets of PDFs; Zahra: upload entire PDFs at once. |
| REQ-010 | Sean, Bradley | Sean: extract text using PyMuPDF; Bradley: convert PDF to raw text for downstream processing. |

## Appendix A - Sean Clayton:

**Prioritization:** MoSCoW prioritization

| Requirement | Rationale | Initial Priority |
|---|---|---|
| The system shall classify each publication as “useful” or “note useful” for predator diet data. | Key requirement for project partner that allows filtering of predator diet publications that used to be a manual process. | Must |
| The system shall extract key predator diet survey data including total individuals, number/proportion of empty stomachs, and predator name. | These fields are critical for determining predator diet trends from publications. | Must |
| The pipeline should accept both single PDFs and folders containing multiple PDFs for batch processing. | Improves efficiency and allows scalability depending on how many PDFs need processing | Should |
| The system shall achieve at least 90% accuracy when classifying “useful” vs. “not useful” papers. | Establishes a concrete measure of success for the model. | Should |
| The extraction process should take less than 2 minutes to complete per PDF. | Improves usability for large scale batch runs of many PDFs. | Could |
| The system shall comply with academic data licensing and operate only on legally obtained publications. | Prevents legal violations when extracting data from publications. | Must |
| The codebase should include automating testing such as unit and integration tests that cover 80% of critical functions. | Ensures that functions and the system itself is behaving as expected to increase accuracy. | Should |
| The system shall automatically apply OCR when a PDF page has no readable text layer. | Many publications are scanned images so OCR is required for PDFs where text cannot be extracted. | Must |
| The system will include a | Essential for handing off the | Must |

| | | |
|---|---|---|
| README that states the process of installing dependencies and running the program. | project to another team or project partner. | |
| The system shall extract text from scientific PDFs using PyMuPDF for accurate, structured text output. | Text from PDF needs to be extracted before passing it along to an LLM. | Must |

## Evidence:

**Problem Statement & Scope**

Sean Clayton **tement:**

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**System Boundaries:**

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- Structured export with provenance and confidence value.
- Reproducible training and evaluation pipelines.

**Out of Scope (Non-Goal):**

- Performing ecological modeling or analysis beyond data extraction.

**Measurable Success Criteria:**

1. **Extraction Accuracy:**

Sean Clayton llback for image based PDFs are all Must Have requirements.

- Should Have requirements add significant value, but could possibly be avoided if needed due to time or budget constraints.
- Could Have items are lower priority enhancements that add minimal value, but would be beneficial if we have time for them.

**Tie Breaking Criteria:**

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| The pipeline should accept both single PDFs and folders containing multiple PDFs for batch processing. | Improves efficiency and allows scalability depending on how many PDFs need processing | Should |
| The system shall achieve at least 90% accuracy when classifying "useful" vs. "not useful" papers. | Establishes a concrete measure of success for the model. | Should |
| The extraction process should take less than 2 minutes to | Improves usability for large scale batch runs of many | Could |

## Appendix B - Raymond Cen

**Prioritization:** MoSCoW

| **Requirement** | **Rationale** | **Initial Priority** |
|---|---|---|
| Allows users to upload a file or multiple PDFs | Needs ability to upload for the model to evaluate and extract. | Must |
| Able to extract specific data from "useful" papers | Needed for structured data, research and analysis. | Must |
| Able to classify paper as "useful" or "not useful" | Part of main functionality of software. | Must |
| Able to provide a confidence score for each classification | Help users see how reliable the classification is. | Must |
| Model is able to achieve >90% accuracy | Ensures that the model is reliable and useful. | Should |
| Should keep a log of processed PDFs and results and errors | Supports debugging and reproducibility. | Should |
| Redact or leave out any personal information from processed PDFs | Protects privacy and complies with data regulations | Should |
| Able to validate that the files uploaded are PDF | Prevents errors and ensures only supported files are processed | Must |
| If multiple PDFs are uploaded it should provide basic stats after processing (number of useful papers, errors, which papers are useful) | Help user quickly see results and understand processed outcomes | Must |
| Able to track time taken to process each PDF | Helps monitor performance and identify bottlenecks | Should |

# Evidence:

**Ethics, Risks, and Constraints Traceability**

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| REQ-002, REQ-006 | Inaccurate or non reproducible extraction of data and values can weaken the trust of the research on the LLM | Maintain a reproducible training and evaluation pipeline. Version datasets, models and configuration. |
| REQ-004, REQ-008 | Inaccessible tools or not enough documentation can exclude less technical researchers and prevent researchers from wanting to use the tool | Provide clear README, documentation, and logs. Test with intended researchers in mind. |
| REQ-004 | Inefficient OCR can inhibit scalability and waste computing resources | Track performance metrics for every PDF. Optimize OCR calls and encourage efficient use of research computing resources |
| REQ-006 | If the training data is skewed then the extracted results can misrepresent global predator and prey patters | Use diverse training sets. Retrain and validate models against underrepresented data categories. Document any known limitations |

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| REQ-005 | Raymond, Zahra, Sean | Raymond: confidence scores |
| REQ-006 | Sean, Raymond, Bradley, Zahra | Sean: academic data licensing compliance; Raymond: redact personal info for privacy; Bradley: adhere to research data regulations and use ML/LLMs; Zahra: follow guidelines and maintain consistency. |
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## Appendix C - Bradley Rule

**Prioritization:** MoSCoW

| Requirement | Rational | Priority |
|---|---|---|
| Able to convert any PDF to a raw text format. | This will be required in order for our program to properly analyze documents, as some PDFs do not easily convert to text. | Must |
| Reliably classify PDFs based on whether they have useful data on the fraction of feeding predators or not. | Our program will need to be able to accurately determine if a document contains the information we are looking for so that we don't overlook useful documents, or waste time processing unusable documents. | Must |
| Scrape APIs and databases for relevant texts | This would reduce the need for individuals to spend time finding documents with potentially useful information. And would likely also drastically speed up the process. | Could |
| Extract the proportion of feeding predators as a variable from documents that have been classified as useful. | This would remove the need for a user to manually search through a document after it has been classified. | Should |
| Utilizes Machine learning and LLM models to perform classification and data extraction | Since this is not a standardized task, we will need to make use of these technologies to achieve our goal. | Must |
| Adherence to regulations and laws pertaining to the acquisition and usage of research articles and data usage. | We need to adhere to laws and regulations regarding the usage and processing of the data we are interested in. | Must |
| Capable of processing large sets of PDF documents | This would streamline the user experience and reduce the need to babysit the program. | Must |
| Able to run unsupervised when analyzing large datasets | Given the large amounts of data this program will need to process, the program should be able to run for long periods of time without human intervention. | Must |

| Requirement | Rational | Priority |
|---|---|---|
| Able to convert any PDF to a raw text format. | This will be required in order for our program to properly analyze documents, as some PDFs do not easily convert to text. | Must |
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| Scrape APIs and databases for relevant texts | This would reduce the need for individuals to spend time finding documents with potentially useful information. And would likely also drastically speed up the process. | Could |
| Able to process figures or models within documents to extract data or assist in classification. | The data we are looking for could be stored in images and models that we could otherwise miss. Processing this properly should help identify and extract information. | Should |
| Output data into spreadsheet format with relevant information for identifying papers and the data they contain. | This will increase the clarity of the output and reduce the need for additional formatting and manual processing to add the information to existing databases. | Should |

**Appendix C - Bradley Rule**

**Prioritization:** MoSCoW

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| Extract the proportion of feeding predators as a variable from documents that have been classified as useful. | This would remove the need for a user to manually search through a document after it has been classified. | Should |
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| Able to process figures or models within documents to extract data or assist in classification. | The data we are looking for could be stored in images and models that we could otherwise miss. Processing this properly should help identify and extract information. | Should |
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## Appendix D - Zahr Alsulaimawi

**Prioritization:** MoSCoW

| Requirement | Rational | Priority |
|---|---|---|
| The system runs efficiently under 5 minutes | Keeps the tools quick to use. | Cloud |
| The system shows error logs | Help user identify failed extractions and any needed debugging | Should |
| System extract data tables from PDFs | Some PDFs have their data in tables. | Must |
| The system should let the user know whether a PDF is useful or not | Saves time by filtering out PDFs with valid needed diet d ata | Must |
| The system should track accuracy . | Prevent the LLM from hallucinating or taking wrong data from PDFs. | Must |
| The system should be able to | Increasing transparency and | Could |

| | | |
|---|---|---|
| let user know what pages from PDFs they got the data | makes validation easier. | |
| The system should follow the guidelines setup | Maintains consistency. | Must |
| The system should be able to to extract data from more than one place in the PDFs | Some pages include results in several locations, not just tables. | Must |
| The system should automatically remove duplicate entities exported data | Prevents redundancy and improves dataset quality. | Should |
| The system should allow user to upload a whole PDF at a time to be analyzed | Simplifies workflow by removing the need for importing only a few pages at a time. | Should |

**Appendix D - Zahr Alsulaimawi**

**Prioritization:** MoSCoW

| Requirement | Rational | Priority |
|---|---|---|
| The system runs efficiently under 5 minutes | Keeps the tools quick to use. | Cloud |
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**Consolidated Requirements**

| ID | Requirement | Acceptance Criteria | Priority |
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| REQ-002 | The system shall extract key predator data( ex. Species, count of empty vs non empty stomach, location and year) | Given useful PDF when the extraction all required fields shall present the output of the data ( ex. Species, count of empty vs non empty stomach, location and year) | Must |
| REQ-003 | The systems shall correctly interpret tables and figures, ensuring the LLM does not hallucinate or takes incorrect values. | Given PDF containing tables or figures, when the table detection process runs, then all values shall match and be flagged if uncertain. | Must |
| REQ-004 | The system shall automatically apply OCR when a PDF page has no readable text layer. | Given an image based or scanned PDF when the system detects no text then OCR shall automatically extract readable text for the further processing. | Must |
| REQ-005 | The system shall track and display accuracy and page numbers for the extracted. | Given an extraction run when the results are outputted then each extracted item shall include page | Must |

| | | | |
|---|---|---|---|
| REQ-006 | The model used for classification and data extraction should be fine tuned for predator-diet surveys. | Given a labeled training dataset when model fine-tuning is completed then the systems classification accuracy shall reach at least 95% and data extraction should reach accuracy 95% when testing valid PDF. | Should |
| REQ-007 | The system shall analyze full PDF inputs to ensure no relevant data sections (tables, captions, appendices, etc.) are missed. | Given a PDF containing multiple sections when extraction module runs then all relevant tables,captions and appendices are scanned and included in the output without anything missed. | Should |
| REQ-008 | The system shall log all processed pages, errors, and any summary statistics. | Given a PDF when the pipeline completes then the system creates a log file that lists total pages processed, any failed pages, errors and success. | Must |
| REQ-009 | The system shall support both single PDF uploads and folders of PDFs. | Given either one PDF or a folder of PDFs when the upload command is executed then the system processes all valid files without user reconfiguration. | Should |
| REQ-010 | The system should be able to convert any PDF to a raw text format | Given any valid PDF file, when the conversion is triggered then a clean readable text version of the document is generated for downstream processing. | Must |